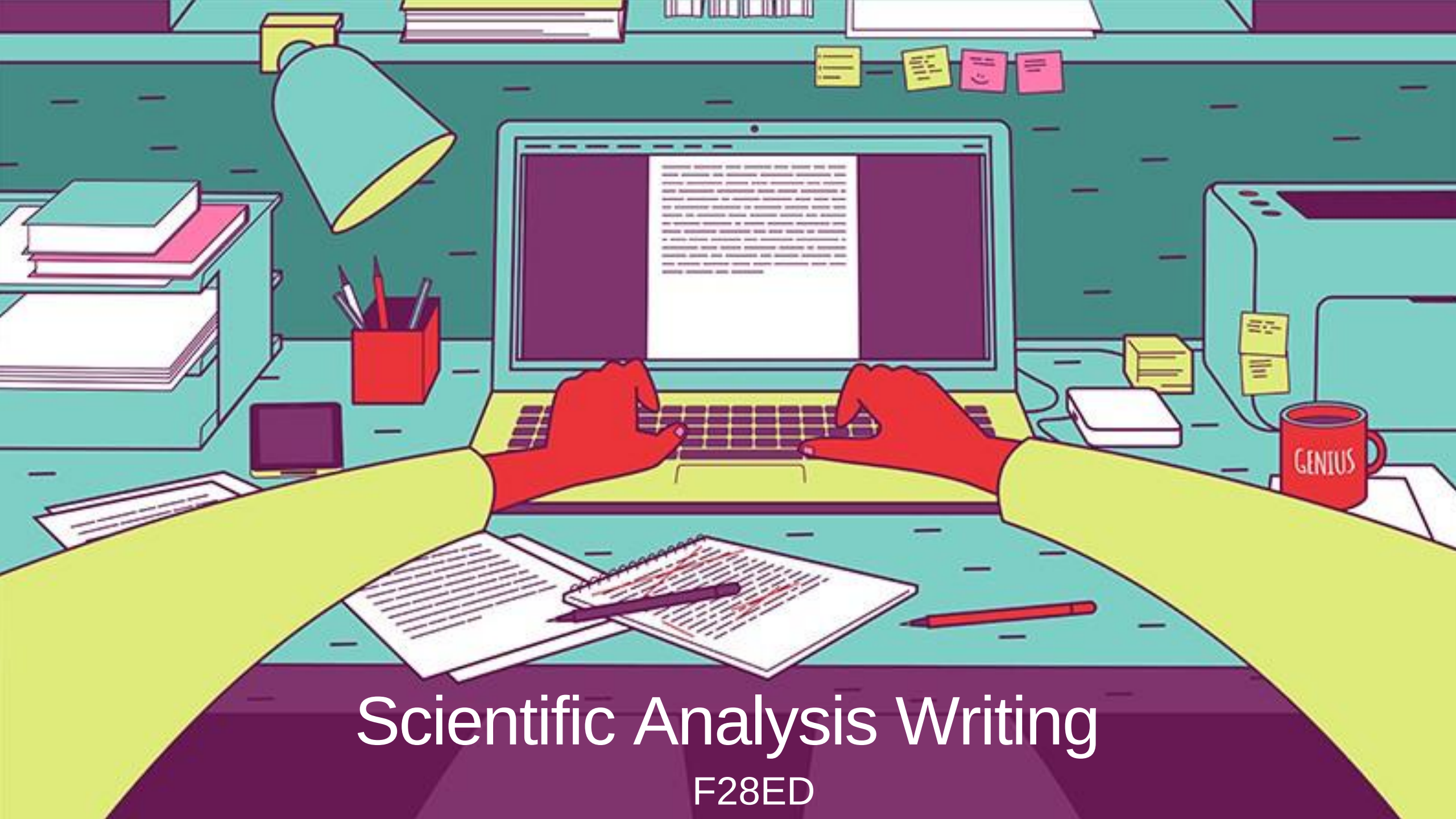


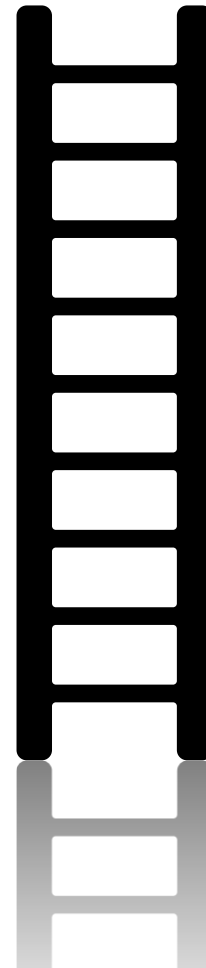


# Scientific Analysis Writing

F28ED

# So far...

- How to work with **users**
- **Design** an experiment
- Define **hypothesis**
- **Conduct** experiment
- **Collect** data
- **Analyse** data



# Today

- Recap about the Robot Embodiment Study
- How to discuss a [real experiment](#)
- Go over [report structure](#)/content
- Look at an [example report](#)

# The Robot Embodiment Study

- Research Question
  - What type of **expressiveness** is appropriate for a **robotic tutor** and which traits are **perceived well** by the users?



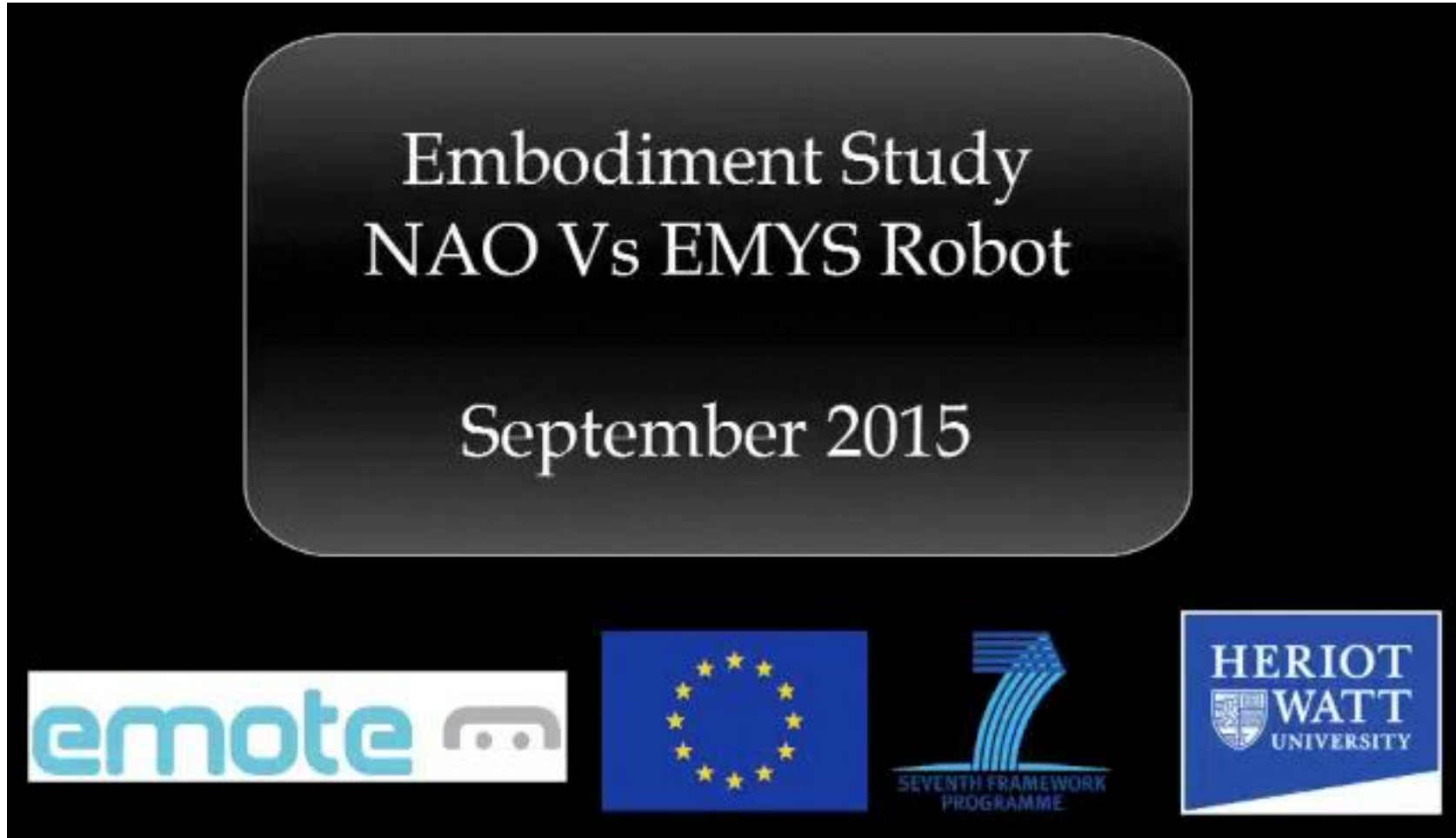
NAO: Can gaze and point with hands



EMYS can gaze and perform facial expressions



# Embodiment Study



[Link To Paper](#)

Search: How Expressiveness of a Robotic Tutor is Perceived by Children in a Learning Environment

# The Robot: Embodiment Study

Participants: 31 students aged between 11-14



NAO: Can gaze and point  
with hands



EMYS can gaze and perform  
facial expressions



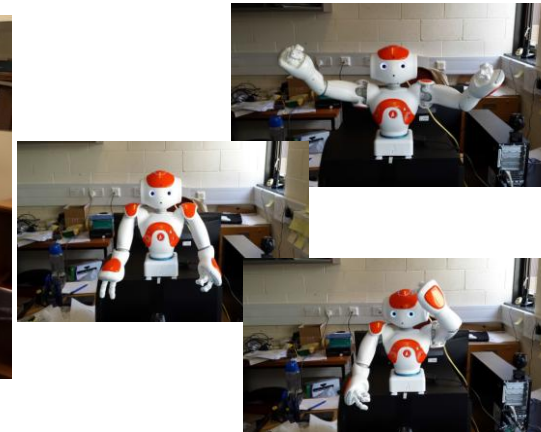


# Between-subjects experiment

## Condition 1 EMYS



## Condition 2 NAO



# Variables

- Independent variable?
  - EMYS vs NAO
- Dependent variables
  - perceived politeness
  - perceived empathy
  - perceived pleasant
  - perceived friendliness
  - time on task

Measured with Semantic Differential response self-report items

## **GODSPEED III: LIKEABILITY**

Please rate your impression of the robot on these scales:

|            |   |   |   |   |   |          |
|------------|---|---|---|---|---|----------|
| Dislike    | 1 | 2 | 3 | 4 | 5 | Like     |
| Unfriendly | 1 | 2 | 3 | 4 | 5 | Friendly |
| Unkind     | 1 | 2 | 3 | 4 | 5 | Kind     |
| Unpleasant | 1 | 2 | 3 | 4 | 5 | Pleasant |
| Awful      | 1 | 2 | 3 | 4 | 5 | Nice     |

“Godspeed”: Another example of a standard questionnaire. (Used in robot studies).

The robot was ...



Adapted with “Smileyometer”



# Hypotheses

- H1: One robot will be **perceived** as **more polite** than the other
  - **Null Hyp**: One robot will **NOT** be perceived as **more polite** than the other
- H2: One robot will be **perceived** as **more friendly** than the other
  - **Null Hyp**: One robot will **NOT** be perceived as **more friendly** than the other
- H3: One robot will be **perceived** as **more empathic** than the other
  - **Null Hyp**: One robot will **NOT** be perceived as **more empathic** than the other
- H4: One robot will be **perceived** as **more pleasant** than the other
  - **Null Hyp**: One robot will **NOT** be perceived as **more pleasant** than the other
- H5: One robot will **enable** the subject to do the task **faster** than the other
  - **Null Hyp**: One robot will **NOT** enable the subject to do the task **faster** than the other

# Time on Task [Exercise]



| DV           | NAO    |    | EMYS   |    | Significance |
|--------------|--------|----|--------|----|--------------|
|              | Mean   | SD | Mean   | SD | p-value      |
| Time on Task | 354sec | 43 | 258sec | 36 | *0.0012      |

Table 2: Results summary for objective measure. SD = Standard deviation. Central tendency for Time on task is mean, DV= Dependent Variable.

- Objective/Subjective?
- Quantitative/Qualitative?
- Nominal/Interval/Ordinal/Ratio?
- T-Test or Mann-Whitney U test?
- Paired or unpaired?
- Significant?

# Time on Task



| DV                  | NAO    |    | EMYS   |    | Significance |
|---------------------|--------|----|--------|----|--------------|
|                     | Mean   | SD | Mean   | SD | p-value      |
| <b>Time on Task</b> | 354sec | 43 | 258sec | 36 | *0.0012      |

Table 2: Results summary for objective measure. SD = Standard deviation. Central tendency for Time on task (ratio) is mean, DV= Dependent Variable. \*indicates a significant value by a unpaired t-test

- Objective/Subjective?
  - Quantitative/Qualitative?
  - Nominal/Interval/Ordinal/Ratio?
  - T-Test or Mann-Whitney U test?
  - Paired or unpaired?
  - Significant?
- Objective
  - Quantitative
  - Ratio
  - T-Test
  - Unpaired
  - Significant!

# Perceived Traits [Exercise]

| DV                | NAO         | EMYS        | Significance |
|-------------------|-------------|-------------|--------------|
|                   | Median/Mode | Median/Mode | p-value      |
| <b>Politeness</b> | 5/5         | 5/5         | .076         |
| <b>Friendly</b>   | 5/5         | 3/4         | .010         |
| <b>Pleasant</b>   | 5/5         | 4/4         | .011         |
| <b>Empathy</b>    | 3/3         | 4/4         | 0.033        |

Table 1: Results summary for subjective measures. Central tendency for Smileometer 5-point rating scale is Median/Mode. DV= Dependent Variable.

- Objective/Subjective?
- Quantitative/Qualitative?
- Nominal/Interval/Ordinal/Ratio?
- T-Test or Mann-Whitney U test?
- Paired or unpaired?
- Significant?

# Perceived Traits



| DV                | NAO         | EMYS        | Significance |
|-------------------|-------------|-------------|--------------|
|                   | Median/Mode | Median/Mode | p-value      |
| <b>Politeness</b> | 5/5         | 5/5         | .076 :(      |
| <b>Friendly</b>   | 5/5         | 3/4         | .010* :-)    |
| <b>Pleasant</b>   | 5/5         | 4/4         | .011* :-)    |
| <b>Empathy</b>    | 3/3         | 4/4         | 0.033* :-)   |

Table 1: Results summary for subjective measures. Central tendency for Smileometer 5-point rating scale (ordinal) is Median/Mode. DV= Dependent Variable. \*indicates a significant value by a unpaired Mann-whitney U test

- Objective/Subjective?
  - Quantitative/Qualitative?
  - Nominal/Interval/Ordinal/Ratio?
  - T-Test or Mann-Whitney U test?
  - Paired or unpaired?
  - Significant?
- Subjective
  - Quantitative
  - Ordinal
  - Mann-Whitney U test
  - Unpaired
  - Friendly/Pleasant/Empathy are significant!

# NULL Hypothesis: Reject or Not Reject?

- H1: One robot will be **perceived** as **more polite** than the other

**p=.076** • Null Hyp: One robot will **NOT** be perceived as **more polite** than the other

Not  
reject

- H2: One robot will be **perceived** as **more friendly** than the other

**p=.010** • Null Hyp: One robot will **NOT** be perceived as **more friendly** than the other

Reject

- H3: One robot will be **perceived** as **more empathic** than the other

**p=.011** • Null Hyp: One robot will **NOT** be perceived as **more empathic** than the other

Reject

- H4: One robot will be **perceived** as **more pleasant** than the other

**p=.0333** • Null Hyp: One robot will **NOT** be perceived as **more pleasant** than the other

Reject

- H5: One robot will **enable** the subject to do the task **faster** than the other

**p=.0012** • Null Hyp: One robot will **NOT** enable the subject to do the task **faster** than the other

Reject



# So what can we say?

- H2: The NAO robot is perceived as significantly more friendly than the EMYS (p< .05 by an unpaired Mann-Whitney U test)
- H3: The NAO robot is perceived as significantly more empathic than the EMYS (p< .05 by an unpaired Mann-Whitney U test)
- H4: The NAO robot is perceived as significantly more pleasant than the EMYS (p< .05 by an unpaired Mann-Whitney U test)
- H5: The EMYS robot enables the user to complete the task in significantly less time than the NAO (p<.05 by an unpaired t-test)

# Subjective qualitative Results

Qualitative questions were asked in a later separate within subjects study.

- Qualitative comments backed up the view that the NAO is overall a better fit as a tutor
- There were 5 negative comments for the NAO compared to 9 for the EMYS e.g., ``..the first [EMYS] robot was a lot more artificial”.
- In addition, there were 15 positive comments for the NAO but only 9 for the EMYS, e.g., ` The [NAO] robot seemed friendly and life-like”.
- More in-depth analysis involves looking at common themes across comments
  - (See previous lecture)

# The Paper: Download and in Pairs Discuss

- Why doesn't the paper discuss time taken?
- Why doesn't the paper discuss politeness? Why do you think this may not have been significant?
- There are two errors in presenting the statistics in this paper. They are subtle and not obvious. What do you think they may be?
- Do you think the results are meaningful? What are the limitations of the work?

# Summary

NAO was found to be significantly more friendly, pleasant and empathetic than EMYS ( $p < 0.05$ , Mann-Whitney unpaired U test), however the EMYS robot helps the user finish the task quicker ( $p < 0.05$  unpaired t-test).

- Plan your experiment analysis. Check it is valid.
- Analysis is there to support a narrative.
- Be careful! Even experienced researchers make errors.